

Getting Started with GreenCloud Simulator

GreenCloud enables the detailed modelling of the energy consumed by a data centre's IT equipment. In this article, the author discusses the installation of GreenCloud, and walks the reader through a few simulation exercises by changing the parameters of the cloud as well as the source code.

GreenCloud is a packet level simulator that uses the existing Network Simulator 2 (NS2) libraries of data centres to track the energy consumed by the different components of a cloud computing environment. It models the various entities of the cloud such as servers, switches, links for communication, and the energy they consume.

It can be helpful in developing solutions to monitor and allocate resources, to schedule workloads for a number of users, optimise the protocols used for communication and also provide solutions for network switches. Data centre upgradation or extension can be decided on using this tool.

NS2 uses two languages, C++ and OTcl (Tool Command Language). The commands from Tcl are usually passed to C++ using the interface, TclCL. GreenCloud uses 80 per cent of the coding done using C++ (TclCL Classes), and the remaining 20 per cent coding is implemented using Tcl scripts (commands are sent from Tcl to C++).

GreenCloud has been developed by the University of Luxembourg and released under the General Public License (GPL).

Installation of GreenCloud

The GreenCloud tool has been developed mainly for Debian-based systems (like Ubuntu, Debian, Linux Mint, etc). The tool will work comfortably with Ubuntu 12.x and later, with kernel version 3.2+. GreenCloud also comes with a pre-configured VM that includes Eclipse to debug NS2, modifying the source code and to start or run simulations.

Here are the instructions for GreenCloud on a non-VM machine. Download the software from this URL: <http://greencloud.gforge.uni.lu/ftp/greencloud-v2.0.0.tar.gz>. Then execute the commands as specified below:

Unzip or untar the software using the command

```
pradeep@localhost $] tar zxvf greencloud-v2.0.0.tar.gz
pradeep@localhost $] cd greencloud-v2.0.0
pradeep@localhost greencloud-v2.0.0 $] ./configure
pradeep@localhost $] ./install-sh
```

(This will install almost as 300MB of software with the dependencies. You need to press "Enter" manually for fewer number of times during the installation, If the installation is unsuccessful, correct the dependencies)
Execute the script by running (This command will pop out a window in a browser with a test simulation data)
pradeep@localhost \$] ./run

Sample simulation

GreenCloud comes with a default test simulation of 144 servers with one cloud user. All the parameters can be varied and tested based on the inputs given to the Tcl file.

The Tcl files are located under the `~greencloud/src/scripts/` directory. There are many scripts that specify the functionality of the cloud environment:

`main.tcl` - specifies the data centre topology and simulation time